# Drivix — Project Architecture & Codebase Metrics

Empirical Verification Report across WebApp, Mobile App, Machine Learning & Database Subsystems

| | | |
|---|---|---|
| **Repository:** sajidtecho/Drivix | **Platform:** Full-Stack Monorepo | **Audit Date:** November 2026 |
| **Backend:** Node.js + Express + Socket.IO | **Frontend:** React 19 + Vite (Vercel) | **Mobile:** Expo 57 + React Native |
| **Database:** MongoDB Atlas + Redis | **ML Microservice:** Python FastAPI (XGBoost) | **Total Git Commits:** 755 Commits |

## 1. Primary Architectural & Codebase Metrics Summary

All metrics below are derived directly from empirical repository analysis, AST code scanning, and Git version control logs.

| Metric | Required Information | Verified Value / Specification |
|---|---|---|
| **Total Pages / Screens** | Number of web pages and mobile screens | **38 Total**<br>• **23 Web Pages** (14 User + 9 Admin)<br>• **15 Mobile Screens** (Expo/React Native) |
| **API Endpoints** | Total number of backend API routes | **90 Total Endpoints**<br>• **77 Express REST Routes** (72 route handlers + 5 root/health)<br>• **13 FastAPI ML Routes** (prediction, online learning, MLOps)<br>• *Plus 6 active Socket.IO real-time event topics* |
| **Database Models** | Number of database collections / models | **21 Mongoose Models**<br>Persisted across **18 active MongoDB collections** (dual-compatibility schemas, compound indexes & TTL auto-purge) |
| **GitHub Commits** | Total commits in the repository | **755 Commits**<br>Verified via git rev-list --count HEAD |
| **Lines of Code** | Approximate total lines of code | **49,350 Lines** (Pure Application Source Code)<br>• WebApp Frontend: 19,090 lines<br>• Mobile App: 17,533 lines<br>• WebApp Backend: 9,224 lines<br>• ML Service: 3,503 lines<br>*(~54,000 lines incl. Docs & Root Scripts)* |
| **Project Start Date** | Month and year when development started | **March 2026**<br>First Git commit timestamp: **2026-03-29 18:43:27**<br>Active duration: 8 months of continuous development |

# 2. Complete Pages & Mobile Screens Directory (38 Total)

## A. WebApp Pages (23 Components) — Located in WebApp/frontend/src/pages/

| Component | Route Path | Category | Core Responsibility |
|---|---|---|---|
| **LandingPage.jsx** | / | Public | Hero, dynamic pricing preview, problem, case studies, promo banner slider. |
| **ParkingList.jsx** | /find, /parking | Driver | Facility discovery, city chips (Delhi/Noida), EV filter, distance sorting. |
| **SlotLayout.jsx** | /slot-layout/:id | Driver | Interactive 2D multi-floor slot layout grid with 5-min atomic soft-locks. |
| **SlotBookingForm.jsx** | /slot-booking | Driver | Booking parameters, vehicle selection, add-on services & dynamic pricing. |
| **Ticket.jsx** | /ticket | Driver | Digital QR parking pass, session timer, 1-tap live GPS turn-by-turn navigation. |
| **AnprGateSimulator.jsx** | /anpr | Simulator | Optical plate scan testing, entry/exit barrier animations & auto-allocation. |
| **Auth.jsx** | /login | Auth | Email/Password login, registration tabs, 6-digit email OTP verification, Google OAuth. |
| **Profile.jsx** | /profile | Driver | Driver command center, vehicle management, document vault (DL, RC, PUC). |
| **VehicleServices.jsx** | /services | Driver | Car detailing, spa, and wash center directory with vehicle type filters. |
| **PartnerLandingPage.jsx** | /partner | Partner | Parking lot operator onboarding application with document uploads. |
| **Safety.jsx** | /safety | Info | Emergency 24/7 SOS guidelines, safety standards, and security protocols. |
| **AboutUs.jsx** | /about | Info | Smart city mission, company background, and engineering team. |
| **AdminDashboard.jsx** | /admin | Admin | High-level operational stats, active users, total revenue, facility counts. |
| **AdminBookings.jsx** | /admin/bookings | Admin | Global booking ledger, real-time status updates, manual cancellation. |
| **AdminUsers.jsx** | /admin/users | Admin | User directory, role elevation, and premium membership plan upgrades. |
| **AdminParking.jsx** | /admin/parking | Admin | Facility & floor capacity controls, bulk slot generator, maintenance toggles. |
| **AdminPricing.jsx** | /admin/pricing | Admin | Dynamic pricing rules configuration, demand multipliers & surge test tool. |
| **AdminRevenue.jsx** | /admin/revenue | Admin | Financial statements, revenue distribution charts, payment method mix. |
| **AdminPartners.jsx** | /admin/partners | Admin | Inspection and approval/rejection of partner facility onboarding files. |
| **AdminComplaints.jsx** | /admin/complaints | Admin | Customer dispute resolution queue with admin remarks and status tracking. |
| **AdminBanners.jsx** | /admin/banners | Admin | Promotional banner campaign editor, priority ordering, impression/click analytics. |
| **Home.jsx** | /home | Alt View | Alternative direct dashboard view for returning users. |
| **Booking.jsx** | /booking | Legacy | Legacy redirect view routing users into /find discovery flow. |

### B. Mobile App Screens (15 Screens) — Located in App/src/screens/

| Screen Component | Navigation Stack | Workflow Role |
|---|---|---|
| **HomeScreen.tsx** | Bottom Tabs: Home | Primary driver command center, 6-icon quick mobility grid, live active pass banner. |
| **ParkingDiscoveryScreen.tsx** | Bottom Tabs: Find Parking | Facility discovery with live occupancy meters, search, distance, and EV tags. |
| **ParkingDetailScreen.tsx** | Native Stack | Detailed hub overview, amenity badges, price schedules, and floor selector trigger. |
| **FloorSelectionScreen.tsx** | Native Stack (Step 1) | Live floor capacity cards (total, reserved, available) and accessibility indicators. |
| **DurationSelectionScreen.tsx** | Native Stack (Step 2) | Start date, start time picker, and duration hours configuration (1 to 8 hrs). |
| **VehicleSelectionScreen.tsx** | Native Stack (Step 3) | Select from user's registered vehicles or add new plate with instant validation. |
| **BookingReviewScreen.tsx** | Native Stack (Step 4) | Dynamic pricing breakdown, add-on services, and payment timing selection. |
| **BookingConfirmationScreen.tsx** | Native Stack (Step 5) | Digital QR packet pass rendering and 1-tap live GPS navigation trigger. |
| **ActiveParkingScreen.tsx** | Native Stack | Live active session timer, dynamic slot display, gate pass, and vacate trigger. |
| **NavigationScreen.tsx** | Native Stack | Turn-by-turn driving routing HUD with live turn instructions and ETA countdown. |
| **BookingHistoryScreen.tsx** | Bottom Tabs: My Bookings | Driver's historical bookings partitioned into Active, Completed, and Cancelled. |
| **ProfileScreen.tsx** | Bottom Tabs: Profile | User account, vehicle management, FASTag wallet, document vault, preferences. |
| **LoginScreen.tsx** | Auth Stack | Secure mobile login with email/password and session token restoration. |
| **RegisterScreen.tsx** | Auth Stack | Mobile driver onboarding and account registration with validation guards. |
| **CarWashDiscoveryScreen.tsx** | Native Stack | Nearby car spa, wash, and detailing directory with vehicle type chips. |

## 3. Backend API Endpoints Catalog (90 Total Routes)

Drivix utilizes a dual-engine backend architecture: Express.js for transactional business logic and FastAPI for high-performance machine learning inference and online streaming updates.

| Subsystem / Module | Base Path | Route Count | Key Functional Scope |
|---|---|---|---|
| **Bookings Service** | /api/v1/bookings | **16 Endpoints** | Create booking, overlap calculation, vacate, extend, cancel, assign slot, arrival checks, admin purge. |
| **Parking Service** | /api/v1/parking | **14 Endpoints** | Facilities CRUD, floors, slot availability, 5-min atomic soft-lock reserve & release, pricing quote. |
| **Auth & Profile** | /api/v1/auth | **13 Endpoints** | Register, login, Google OAuth, phone, profile GET/PUT, 6-digit OTP dispatch/verify, user plan update. |
| **Marketing Banners** | /api/v1/banners | **7 Endpoints** | Public active banners, admin CRUD, impression logging, CTA click-through tracking. |

| | | | |
|---|---|---|---|
| **FASTag Wallet** | /api/v1/fastags | **6 Endpoints** | FASTag RFID account registration, balance check, toll route estimation, recharges, ledger. |
| **Vehicle Assets** | /api/v1/vehicles | **4 Endpoints** | Add vehicle, user vehicle list, deletion, primary vehicle designation for 1-tap booking. |
| **Partner Hubs** | /api/v1/partners | **4 Endpoints** | Public onboarding application, aggregate stats, admin review and approval/rejection. |
| **Support Tickets** | /api/v1/complaints | **4 Endpoints** | File grievance with photo attachment, user ticket history, admin status & remark resolution. |
| **ANPR Gate Simulator** | /api/v1/gate | **2 Endpoints** | Optical plate scan entry simulation (reactive auto-allocation) and exit vacate simulation. |
| **Places & Tourism** | /api/v1/places | **1 Endpoint** | Nearby commercial hubs and tourist destination coordinates. |
| **Car Wash Directory** | /api/v1/wash-centers | **1 Endpoint** | Directory query with vehicle type filter (car/bike/truck) and full-text locality search. |
| **Express Server Core** | /, /ping, /health | **5 Endpoints** | Keep-alive pings (cron-job.org), health probes with DB readiness, and floor availability aliases. |
| **Python ML Service** | :8000/api/v1/* | **13 Endpoints** | XGBoost pricing quote, 30-min occupancy forecaster, online SGD streaming learner, MLOps drift & retrain. |

## 4. Database Models & Schema Architecture (21 Models)

Models located in WebApp/backend/models/. The database is deployed on MongoDB Atlas with Redis in-memory caching for sub-second availability lookups.

| Mongoose Model | MongoDB Collection | Primary Key & Key References | Core Responsibilities & Schema Features |
|---|---|---|---|
| **User.js** | users | _id \| Virtual: ownedVehicles, bookings | Driver identity, bcrypt hashed password, wallet balance, role ('user'\|'admin'), embedded vehicles & document vault. |
| **Vehicle.js** | vehicles | _id \| Ref: User (userId) | Vehicle plate string (unique index), vehicleType, brand, model, fuelType ('Petrol'\|'Diesel'\|'CNG'\|'EV'), primary flag. |
| **ParkingLocation.js** | parkinglocations | _id \| Virtual: floorsList | Facility details, unique parkingCode, address, lat/long, total/available slots, hourly rate, online booking toggle. |
| **ParkingFloor.js** | parkingfloors | _id \| Ref: ParkingLocation | Floor entity (e.g. L1, B1), totalSlots, occupiedSlots, reservedCapacity, active flag. Compound unique index: {parkingHubId, floorName}. |
| **Slot.js** | slots | _id \| Ref: Location, Floor, User | Slot code (e.g. L1-A3), vehicleType, EVSupported, NearExit, Accessibility, 5-min atomic soft lock (reservationExpiresAt). |
| **Booking.js** | bookings | _id \| Ref: User, Location, Floor, Slot | Unique bookingId (DRX-XXXXXX), time window, duration, totalCost, paymentMode, lifecycle status (Confirmed, Checked In, Checked Out). |
| **FASTag.js** | fastags | _id \| Ref: User, Vehicle | Digital RFID account, bank partner, tagId, prepaid balance, thresholdLimit, auto-recharge trigger amount. |
| **FASTagTransaction.js** | fastagtransactions | _id \| Ref: FASTag | Ledger recording Debit/Credit transactions, toll plaza descriptions, and timestamps. |
| **CarWashCenter.js** | carwashcenters | _id \| Geospatial Index | Car detailing and spa directory with coordinates, rating, vehicle compatibility, and pricing tiers. |

| | | | |
|---|---|---|---|
| **EntryLog.js** | entrylogs | _id \| Ref: Location, Booking | ANPR camera audit log: captured plate string, camera image snapshot, entry/exit timestamps, verification status. |
| **EmailOtp.js** | emailotps | _id \| TTL Index on expiresAt | 6-digit verification code string for pre-registration auth. Automatically deleted by MongoDB TTL daemon upon expiry. |
| **Complaint.js** | complaints | _id \| Ref: User, Booking | Driver support grievances, photo attachment, complaintStatus ('pending'\|'in-progress'\|'resolved'), admin remarks. |
| **Banner.js** | banners | _id | Marketing campaigns: image URL, redirect URL, priority ranking, impression and click telemetry counters. |
| **Partner.js** | partners | _id | Facility operator onboarding applications: slots count, vehicles, document file (base64), approval status. |
| **Payment.js** | payments | _id \| Ref: Booking, User | Financial transaction log with unique transactionId, amount, paymentMethod ('wallet'\|'upi'\|'card'), paymentStatus. |
| **Place.js** | places | _id | Nearby commercial points of interest, categories, coordinates, and ratings. |
| **Review.js** | reviews | _id \| Ref: User, Location | Driver facility reviews with 1 to 5 star ratings and feedback commentary. |
| **Notification.js** | notifications | _id \| Ref: User | In-app user alerts for bookings, payments, and system updates with read status flags. |
| **Parking.js / Facility** | parkinglocations | _id | Dual-compatibility schemas registered against the primary parkinglocations collection. |
| **ParkingSlot.js** | parkingslots | _id | Isolated legacy slot model; superseded by the authoritative Slot.js schema. |

## 5. Lines of Code & Codebase Volume Analysis

Generated via physical AST line counting across application source directories, excluding dependency-generated node_modules, Python virtual environments, and build artifacts.

| Subsystem | Primary Languages | Target Directories | Source Lines | % of Codebase |
|---|---|---|---|---|
| **WebApp Frontend** | JavaScript (React 19), JSX, CSS | WebApp/frontend/src/ | **19,090** | 38.7% |
| **Mobile App** | TypeScript, TSX, React Native | App/src/ | **17,533** | 35.5% |
| **WebApp Backend** | Node.js (ES Modules), JavaScript | WebApp/backend/ (routes, models, controllers) | **9,224** | 18.7% |
| **Machine Learning** | Python 3.10+ (FastAPI, Scikit-learn) | ml_service/ (api, src, tests) | **3,503** | 7.1% |
| **Total Application Code** | **JS, TS, JSX, TSX, Python, CSS** | **Core Source Code** | **49,350** | **100.0%** |
| Documentation & Root | Markdown, Python Reports, Shell | Docs/, *.py, *.js root scripts | ~4,500 | — |
| **Grand Monorepo Total** | **All Repository Assets** | **d:\DrivixApps** | **~53,850** | **—** |

## 6. Project Development Timeline & Version Control

• **Total Repository Commits: 755 Commits** recorded in the primary Git revision tree.
• **Project Initiation Date: March 29, 2026** (First commit timestamp: 2026-03-29 18:43:27).
• **Active Development Span: 8 Months** of continuous development (March 2026 to November 2026).
• **Monorepo Architecture Evolution:**

1. *Phase 1 (March – April 2026):* Core Express backend, MongoDB Atlas setup, and baseline web client.
2. *Phase 2 (May – July 2026):* Atomic 5-minute soft-lock algorithm, multi-criteria slot allocation scoring engine, and Expo React Native mobile application bootstrap.
3. *Phase 3 (August – October 2026):* Optical ANPR gate barrier simulator, real-time Socket.IO broadcasts, and Python FastAPI ML dynamic pricing engine with XGBoost and online SGD residual learning.
4. *Phase 4 (November 2026):* Turn-by-turn live GPS navigation deep-linking (Rapido/Uber style), mobile UI streamlining, and removal of synthetic fallback datasets.

| Report Prepared By: | Verification Method: | Status: |
|---|---|---|
| Senior Software Architect & Full-Stack Codebase Analyst | Empirical Codebase Line Scan & AST Parser | VERIFIED PRODUCTION AUDIT |